

Convergence of Monte Carlo Optimistic Policy Iteration: Statewise Cumulative Learning and Sharpness

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Abstract

We prove almost-sure convergence of initial-visit Monte Carlo optimistic policy iteration when initial actions are uniform within each state and every state accumulates infinite learning weight. State sampling may be adaptive and may vanish for arbitrarily long intervals. The result covers finite discounted and proper undiscounted episodic Markov decision processes, unbiased returns with uniformly bounded conditional variance, fixed-priority or fully inertial ties, and square-summable, eventually nonincreasing deterministic steps, without an additional comparability condition. The estimates converge to the optimal action values, the policy-evaluation target changes only finitely often, and all sufficiently late policies are optimal. Three sampled counterexamples show that convergence need not survive replacing Monte Carlo targets by forward TD(0.99), updating every first-visited pair, or biasing initial actions toward the greedy action. Each changes only one of these algorithmic choices and retains a positive sampling floor. The examples establish sharpness in these three directions, without asserting absolute maximality.

Keywords: reinforcement learning, optimistic policy iteration, Monte Carlo exploring starts, stochastic approximation, policy improvement

1 Introduction

Policy iteration alternates between evaluating a policy and improving it using the resulting value estimates. In its optimistic form, the policy is improved before evaluation has converged. Monte Carlo Exploring Starts (MCES), also called Monte Carlo optimistic policy iteration (MC-O-PI), is an example: an episode starts from a sampled state–action pair, follows a greedy policy, and supplies a return used to update the action-value estimate. The policy is then selected greedily again. A small change in the estimates near a tie can change the policy, and therefore the target of the next Monte Carlo update.

The convergence question has a long history (Sutton, 1999). Tsitsiklis (2002) established convergence for discounted optimistic policy iteration with uniform updates, and Chen (2018) and Liu (2021) studied extensions to stochastic shortest path problems. These works also distinguish sampling frequencies from effective update rates: component-dependent step sizes can compensate for nonuniform sampling. A different line of work obtains convergence from structural assumptions on the MDP, such as an optimal policy feed-forward structure (Wang

et al., 2022). These results concern different combinations of update rules and assumptions; they do not make initial-visit, first-visit, and every-visit versions interchangeable.

This paper studies initial-visit MCES, in which only the initial state–action pair of each episode is updated. A common step size suffices when initial action probabilities are equal within each state and each state accumulates infinite learning weight:

$$\sigma_k(s, a) = \sigma_k(s) \geq 0, \quad \sum_{k=0}^{\infty} \alpha_k \sigma_k(s) = \infty \quad (s \in \mathcal{S}).$$

The probability $\sigma_k(s)$ is per action. State probabilities may differ, depend on the entire history, and be zero for arbitrarily long intervals. The update does not divide by the sampling probability. Infinite visits alone are insufficient as a learning condition: the steps attached to those visits must have divergent sum.

Theorem 4 covers finite discounted MDPs and proper undiscounted episodic MDPs, with the return moments and steps specified in Section 2.2. It permits fixed-priority ties or full inertia, which retains a selected action whenever it remains greedy. Almost surely, $q_k \rightarrow q^*$, the target q_{π_k} changes only finitely often, and all sufficiently late policies are optimal. Policy labels need not stabilize when optimal actions are tied.

Proposition 12 gives three sampled counterexamples, each changing one algorithmic choice: the Monte Carlo target, the initial-pair-only update, or equality of initial action probabilities within a state. Each retains fixed-priority ties, bounded sampled update targets, a positive sampling floor, and a common nonincreasing Robbins–Monro step sequence. Thus their failures do not exploit weak state exploration. All three have nonvanishing oscillations of the optimal-action estimates.

The convergence proof uses the restoring drift of policy-relative gaps. Favorable updates create positive gaps, a statewise martingale barrier protects them, and a finite ordering of policy targets forces eventual stabilization. For the counterexamples, robust returns between separated sets persist on one infinite-horizon noise event.

Section 2 specifies the MDP and algorithm, and Section 3 proves convergence. Section 4 presents the three constructions and Section 5 states their scope. The nonconvergence proofs and quantitative bounds are in Appendices A–D; Appendix E describes the certificates and finite numerical illustrations.

2 Preliminaries and the MCES algorithm

2.1 Finite discounted and proper episodic MDPs

Let $\mathcal{S}$ be a finite set of nonterminal states, with finite nonempty action sets $\mathcal{A}(s)$, and put $\mathcal{I} = \{(s, a) : s \in \mathcal{S}, a \in \mathcal{A}(s)\}$ and $J = |\mathcal{I}|$. A transition from (s, a) produces a reward and a next state according to the MDP kernel. Expected absolute one-step rewards are finite. Terminal states, when present, have value zero and are omitted from $\mathcal{I}$. Write $\mathcal{P}$ for the finite set of deterministic stationary policies.

The discount factor is $\gamma \in [0, 1]$. If $\gamma = 1$, assume that every $\pi \in \mathcal{P}$ is *proper*: from every nonterminal state it reaches a terminal state almost surely. If $\gamma < 1$, no termination

assumption is imposed. State and action values are

$$v_\pi(s) = \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t R_t \mid S_0 = s \right], \quad q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t R_t \mid S_0 = s, A_0 = a \right],$$

where rewards after termination are zero and the initial action in the second expression is prescribed. The policy is followed thereafter. The values are finite: discounting suffices when $\gamma < 1$, while properness makes each finite nonterminal transition matrix transient when $\gamma = 1$. In particular,

$$B := \max_{\pi \in \mathcal{P}} \|q_\pi\|_\infty < \infty. \quad (1)$$

All vector inequalities below are coordinatewise, and all norms are supremum norms unless indicated otherwise. We use $\inf \emptyset = \infty$.

Lemma 1 (Uniform termination in the episodic case) *If every deterministic stationary policy is proper, there are an integer $m \geq 1$ and $\rho \in [0, 1)$ such that the termination time T satisfies $\mathbb{P}(T > jm) \leq \rho^j$ under every control law, uniformly over starting states and prescribed initial actions. In particular, $\sup \mathbb{E}[T^2] < \infty$.*

Proof. Let $w_n(s)$ be the largest probability of remaining nonterminal after n transitions, allowing arbitrary history-dependent policies. Then $w_0(s) = 1$ and finite-horizon dynamic programming gives

$$w_{n+1}(s) = \max_{a \in \mathcal{A}(s)} \sum_{s' \in \mathcal{S}} P(s' \mid s, a) w_n(s').$$

The sequence decreases to a vector w . Finiteness permits passage through the maximum and sum. A deterministic stationary maximizer π therefore satisfies $w = P_\pi w$. Properness implies $P_\pi^n \rightarrow 0$, so $w = 0$. Choose m with $\max_s w_m(s) \leq \rho < 1$. Conditioning at successive blocks of m transitions gives the asserted bound, also when the first action is prescribed. Summing the geometric tail gives a uniform second moment. $\blacksquare$

For a deterministic policy ρ , write $T_\rho v = r_\rho + \gamma P_\rho v$, and let $Tv(s) = \max_a \{r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) v(s')\}$. The following form of policy improvement will be used repeatedly; see also Howard (1960).

Lemma 2 (Policy improvement and Bellman verification) *Suppose that, for every state,*

$$q_\pi(s, \rho(s)) \geq q_\pi(s, \pi(s)).$$

Then $v_\rho \geq v_\pi$ and $q_\rho \geq q_\pi$. If additionally $q_\rho \neq q_\pi$, then $\sum_{i \in \mathcal{I}} q_\rho(i) > \sum_{i \in \mathcal{I}} q_\pi(i)$. A deterministic policy greedy for its own action values is optimal, including among history-dependent policies.

Proof. The premise is $T_\rho v_\pi \geq v_\pi$. Since $v_\rho = T_\rho v_\rho$,

$$v_\rho - v_\pi = (I - \gamma P_\rho)^{-1} (T_\rho v_\pi - v_\pi) \geq 0. \quad (2)$$

The inverse is the nonnegative series $\sum_{j \geq 0} (\gamma P_\rho)^j$: discounting or properness ensures convergence. The identity $q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) v_\pi(s')$ gives the action-value comparison. Distinct coordinatewise ordered vectors have strictly ordered sums.

If π is greedy for q_π , then $Tv_\pi = v_\pi$. Iterating the Bellman inequality along any control law shows that its expected return is at most v_π , up to a terminal remainder bounded by $\|v_\pi\|_\infty \gamma^n$ when $\gamma < 1$, or by $\|v_\pi\|_\infty \mathbb{P}(T > n)$ when $\gamma = 1$. That remainder tends to zero, in the latter case by Lemma 1. The policy π attains v_π , proving optimality. ■

An optimal deterministic stationary policy exists. For example, choose π maximizing $\sum_s v_\pi(s)$ over the finite set $\mathcal{P}$. If it were not greedy for q_π , a greedy improvement ρ would have $T_\rho v_\pi - v_\pi \geq 0$ with a positive coordinate; the identity term in the inverse in (2) would give a strict increase of the sum, a contradiction. We denote the optimal state and action values by v^* and q^* , respectively. We do not use the converse implication from action-value ordering to state-value ordering. Equal action-value targets can belong to policies with different state values.

2.2 Initial-visit Monte Carlo control

Let $q_k \in \mathbb{R}^{\mathcal{I}}$ be the current estimate, starting from a deterministic finite q_0 . At the beginning of iteration k , select a deterministic greedy policy

$$\pi_k \in \mathcal{G}(q_k), \quad \mathcal{G}(q) = \{\pi \in \mathcal{P} : q(s, \pi(s)) = \max_a q(s, a) \text{ for all } s\}. \quad (3)$$

Let $\mathcal{H}_k$ contain the history after computing q_k but before this selection. We allow either of two tie rules, fixed throughout a run:

1. *Fixed priorities.* At every state use a fixed priority ordering of the actions, including at initialization.
2. *Full inertia.* Choose π_0 greedily, and thereafter retain the preceding action whenever it remains greedy:

$$\pi_{k+1}(s) = \pi_k(s) \quad \text{if } \pi_k(s) \in \arg \max_a q_{k+1}(s, a). \quad (4)$$

Otherwise choose any maximizing action, possibly at random and using the history. No statewise independence is required for this rule.

After selecting π_k , choose the initial-pair distribution σ_k and let $\mathcal{F}_k$ contain $\mathcal{H}_k$, π_k , and σ_k . The distribution may depend on the policy and on the entire past. Conditional on $\mathcal{F}_k$, draw I_{k+1} with distribution σ_k and obtain a return estimate G_{k+1} . In the episodic implementation, start with that pair, follow π_k thereafter until termination, and sum the discounted rewards. Only the initial pair is updated:

$$q_{k+1}(i) = q_k(i) + \alpha_k \mathbf{1}_{\{I_{k+1}=i\}}(G_{k+1} - q_k(i)), \quad i \in \mathcal{I}. \quad (5)$$

The policy remains fixed during each episode and is selected again after this update. The next pre-policy history contains the pair, return, and update; in particular,

$$\mathcal{H}_k \subseteq \mathcal{F}_k \subseteq \sigma(\mathcal{F}_k, I_{k+1}) \subseteq \mathcal{H}_{k+1} \subseteq \mathcal{F}_{k+1}.$$

All stopping times in the convergence proof refer to $(\mathcal{F}_k)$. This is the initial-visit version of MC-O-PI, also called Monte Carlo control with exploring starts (MCES). First-visit and every-visit algorithms update additional pairs within an episode; those algorithms are not covered by the convergence theorem below.

Assumption 3 (Cumulative learning, returns, and step sizes) *The following conditions hold:*

- (1) *Almost surely, initial actions are uniform within each state, and every state accumulates infinite learning:*

$$\begin{aligned} \sigma_k(s, a) = \sigma_k(s) \geq 0, \quad \sum_s |\mathcal{A}(s)| \sigma_k(s) = 1, \quad k \geq 0, \\ \sum_{k=0}^{\infty} \alpha_k \sigma_k(s) = \infty \quad \text{for every } s. \end{aligned} \quad (6)$$

Here $\sigma_k(s)$ is the probability of each pair at state s , not its state marginal, which is $|\mathcal{A}(s)| \sigma_k(s)$.

- (2) *For a deterministic $c_v < \infty$, the return is conditionally unbiased with uniformly bounded conditional variance. On pairs with positive conditional sampling probability,*

$$\begin{aligned} \mathbb{E}[G_{k+1} \mid \mathcal{F}_k, I_{k+1} = i] &= q_{\pi_k}(i), \\ \mathbb{E}[(G_{k+1} - q_{\pi_k}(i))^2 \mid \mathcal{F}_k, I_{k+1} = i] &\leq c_v. \end{aligned} \quad (7)$$

No return law is required on a zero-probability initial-pair event.

- (3) *The deterministic steps satisfy*

$$0 < \alpha_k \leq 1, \quad \sum_{k=0}^{\infty} \alpha_k^2 < \infty, \quad \alpha_{k+1} \leq \alpha_k \quad (k \geq k_m), \quad (8)$$

for a deterministic integer k_m . Thus nonincrease is required only eventually. Equation (6) already implies $\sum_k \alpha_k = \infty$.

The state law can be unequal, adaptive, and zero for arbitrarily long intervals. Equivalently, draw a state with probability $d_k(s)$ and then choose its initial action uniformly, giving $\sigma_k(s) = d_k(s)/|\mathcal{A}(s)|$. No lower bound or bound on waiting times is imposed in Assumption 3. Infinite visitation alone is not a substitute for infinite *learning weight* when a common global step sequence is used.

Bounded one-step rewards suffice for (7) with ordinary episode returns. For $|R_t| \leq R_{\max}$ and $\gamma < 1$, the return is bounded by $R_{\max}/(1 - \gamma)$. For $\gamma = 1$ it is bounded by $R_{\max}T$, which has a uniform second moment by Lemma 1. Unbounded rewards or other estimators are allowed when the explicit conditional moments hold. In a continuing discounted problem, an infinite rollout is unnecessary: with bounded rewards, draw an independent U with $\mathbb{P}(U = j) = (1 - \gamma)\gamma^j$, simulate from the prescribed initial pair using π_k thereafter, and return $R_U/(1 - \gamma)$. This bounded estimator is unbiased; for $\gamma = 0$ take $U = 0$.

No comparability condition such as $\alpha_{k+j} \geq c\alpha_k$ for j up to order $1/\alpha_k$ is assumed. Nonincrease compares earlier seed updates with a state's eventual completion step and controls future variance on that state's learning clock.

3 Results

Theorem 4 (Convergence under cumulative learning) *Consider recursion (5) under Assumption 3, with either fixed-priority or full-inertia tie-breaking from Section 2.2. Then, almost surely, (q_{π_k}) changes only finitely many times and*

$$q_k \longrightarrow q^*.$$

Every sufficiently late selected policy π_k is optimal. If each state has a unique optimal action, π_k is eventually constant; with optimal ties, constancy of policy labels is not asserted.

3.1 The mean-field argument

Replace each sampled update by its conditional expectation:

$$q_{k+1}(s, a) = q_k(s, a) + \alpha_k \sigma_k(s) (q_{\pi_k}(s, a) - q_k(s, a)). \quad (9)$$

At a state with $c = \alpha_k \sigma_k(s) > 0$, greediness before and after the update gives

$$\begin{aligned} q_{\pi_k}(s, \pi_{k+1}(s)) - q_{\pi_k}(s, \pi_k(s)) &\geq \frac{1-c}{c} (q_k(s, \pi_k(s)) - q_k(s, \pi_{k+1}(s))) \\ &\geq 0. \end{aligned}$$

If $c = 0$, the state's table is unchanged. Fixed priorities and full inertia retain its selected action, so the same policy-improvement inequality holds with equality. Thus Lemma 2 gives $q_{\pi_{k+1}} \geq q_{\pi_k}$. With finitely many targets, this sequence is eventually a single $\bar{q}$. Infinite cumulative learning then makes (9) converge to $\bar{q}$. A policy $\hat{\pi}$ occurring infinitely often is greedy for $\bar{q} = q_{\hat{\pi}}$, so Bellman verification gives $\bar{q} = q^*$.

Equal action rates are exactly what permits a common coefficient c . The sampled recursion need not improve its target at every change. Instead we will show a uniform positive conditional probability of improving or stopping during each sufficiently late target interval.

3.2 Sampling, filtering, and stability

Lemma 5 (From conditional to actual update weight) *Let $J_{k+1}(i) \in \{0, 1\}$ be $\mathcal{F}_{k+1}$ -measurable indicators for finitely many coordinates, and put $p_k(i) = \mathbb{E}[J_{k+1}(i) \mid \mathcal{F}_k]$. If the deterministic $\alpha_k \geq 0$ satisfy $\sum_k \alpha_k^2 < \infty$, then on one probability-one event, for every i ,*

$$\sum_k \alpha_k p_k(i) = \infty \implies \sum_k \alpha_k J_{k+1}(i) = \infty. \quad (10)$$

This statement does not require an assumption on the update targets.

Proof. The series $\sum_k \alpha_k (J_{k+1}(i) - p_k(i))$ is a martingale series whose increment variances sum to at most $\sum_k \alpha_k^2$. It therefore converges almost surely by the L^2 martingale convergence theorem. Subtracting its partial sums from $\sum_{k \leq n} \alpha_k J_{k+1}(i)$ proves the implication. Intersect the probability-one events over the finitely many coordinates. $\blacksquare$

For initial-pair indicators, $p_k(i) = \sigma_k(i)$. In particular, a positive initial-pair floor and $\sum_k \alpha_k = \infty$ give infinite actual update weight. A full first-visit indicator dominates the initial-pair indicator, so the same conclusion holds for full first visits.

Lemma 6 (Filtering accumulated errors) *Consider the scalar recursion*

$$x_{k+1} = (1 - b_k)x_k + b_k y_k + e_{k+1}, \quad 0 \leq b_k \leq 1.$$

For $n > r$, its filtered error satisfies

$$\left| \sum_{k=r}^{n-1} e_{k+1} \prod_{j=k+1}^{n-1} (1 - b_j) \right| \leq 2 \max_{r \leq m \leq n} \left| \sum_{k=r}^{m-1} e_{k+1} \right|. \quad (11)$$

If $\sum_k b_k = \infty$ and $\sum_k e_{k+1}$ converges, eventual bounds $\ell \leq y_k \leq h$ imply $\ell \leq \liminf_k x_k \leq \limsup_k x_k \leq h$. In particular, an eventually constant target y gives $x_k \rightarrow y$. All statements are pathwise.

Proof. Unrolling from r gives the initial value with weight $P_{r,n} = \prod_{k=r}^{n-1} (1 - b_k)$, the targets with nonnegative weights summing to $1 - P_{r,n}$, and the displayed error. The error weights belong to $[0, 1]$ and are nondecreasing in k , so summation by parts gives (11). If $\sum_k b_k = \infty$, then $P_{r,n} \rightarrow 0$ for each r . Convergence of the error series makes its tail partial sums uniformly small. First let $n \rightarrow \infty$ with r after the target bounds begin, and then let $r \rightarrow \infty$. ■

Lemma 7 (Stability and persistent targets) *Under Assumption 3, with any greedy tie selection, there is one probability-one event on which*

$$\sum_k \alpha_k \mathbf{1}_{\{I_{k+1}=i\}} = \infty \quad (i \in \mathcal{I}), \quad \limsup_k \|q_k\|_\infty \leq B.$$

On the same event, if $q_{\pi_k} = \bar{q}$ for all sufficiently large k , then $q_k \rightarrow \bar{q} = q^$.*

Proof. For $i = (s, a)$, put

$$\lambda_{k,i} = \alpha_k \mathbf{1}_{\{I_{k+1}=i\}}, \quad \varepsilon_{k+1,i} = \lambda_{k,i} (G_{k+1} - q_{\pi_k}(i)).$$

Cumulative learning and Lemma 5 give $\sum_k \lambda_{k,i} = \infty$ almost surely. The return-moment assumption makes $\varepsilon_{k+1,i}$ a martingale difference with conditional second moment at most $c_v \alpha_k^2 \sigma_k(s)$. Its series therefore converges almost surely. Intersect these events over the finitely many coordinates.

On this event the MCES recursion has the form of Lemma 6, with weights $\lambda_{k,i}$, targets $q_{\pi_k}(i) \in [-B, B]$, and errors $\varepsilon_{k+1,i}$. The lemma gives stability and convergence to any eventual target $\bar{q}$. A policy $\hat{\pi}$ occurring infinitely often satisfies $q_{\hat{\pi}} = \bar{q}$, and its greedy inequalities pass to the limit. Lemma 2 gives $\bar{q} = q^*$. No conditioning on eventual target persistence is used. ■

3.3 A barrier on a state's learning clock

The same sampling factor appears in a gap's drift and variance. Keeping that factor permits arbitrarily long pauses: when a state is not sampled, both the restoring force and the noise pause. No divergence of the state's clock is needed for the following local estimate.

Lemma 8 (State-clock barrier) *Let (D_k) be a real adapted process with integrable increments, let $r \leq \tau$ be stopping times, and let $\sigma_k(s) \in [0, 1]$ be $\mathcal{F}_k$ -measurable. The deterministic steps are in $(0, 1]$, square summable, and nonincreasing from r onward. Write*

$$m_k = \mathbb{E}[D_{k+1} - D_k \mid \mathcal{F}_k], \quad Z_{k+1} = D_{k+1} - D_k - m_k.$$

Suppose deterministic $\delta, V > 0$ satisfy, on $\{r \leq k < \tau, D_k \geq 0\}$,

$$m_k \geq \alpha_k \sigma_k(s)(4\delta - D_k), \quad \mathbb{E}[Z_{k+1}^2 \mid \mathcal{F}_k] \leq V \alpha_k^2 \sigma_k(s). \quad (12)$$

Then, on $\{r < \infty, D_r > 0\}$,

$$\mathbb{P}(\exists \text{ finite } n \in [r, \tau] : D_n \leq 0 \mid \mathcal{F}_r) \leq C_0 \frac{\alpha_r}{\min\{D_r, \delta\}} + \frac{C_1}{\delta^2} \sum_{k=r}^{\infty} \alpha_k^2, \quad (13)$$

where $C_0 = \max\{1, 20V(1 + (3\delta)^{-1})\}$ and $C_1 = 16V$. The crossing event includes the update ending at τ .

Proof. *Before reaching δ .* Suppose $0 < D_r < \delta$. If $D_r < \alpha_r$, the first term in (13) is already at least one. Otherwise define

$$\nu = \tau \wedge \inf\{n \geq r : D_n \leq 0 \text{ or } D_n \geq \delta\}, \quad M_n = \sum_{k=r}^{n-1} \mathbf{1}_{\{k < \nu\}} Z_{k+1},$$

and the state clock

$$A_s(r, n) = \sum_{k=r}^{n-1} \alpha_k \sigma_k(s). \quad (14)$$

A first crossing at $n \leq \tau$ before reaching δ forces

$$M_n \leq -D_r - 3\delta A_s(r, n). \quad (15)$$

For $b > 0$ known at r , set $T(b) = \inf\{n \geq r : A_s(r, n) \geq b\}$, possibly infinite. Nonincrease bounds the overshoot and future variance, at every finite horizon $n \geq r$, by

$$A_s(r, T(b) \wedge n) \leq b + \alpha_r, \quad \sum_{k=r}^{T(b) \wedge n-1} \alpha_k^2 \sigma_k(s) \leq \alpha_r(b + \alpha_r).$$

Conditional martingale isometry and Doob's L^2 inequality therefore give

$$\mathbb{P}\left(\sup_{r \leq n \leq T(b)} |M_n| \geq z \mid \mathcal{F}_r\right) \leq \frac{4V \alpha_r(b + \alpha_r)}{z^2}, \quad z > 0. \quad (16)$$

The supremum is over finite indices. First stop at a finite horizon, then increase the horizon; an unattained supremum is handled by increasing thresholds to z from below. For $\mathcal{F}_r$ -measurable b, z , localize on sets where b, z, z^{-1} are bounded. Random r is treated on $\{r = j\}$ and then combined over integers j . Thus no finite bound on $T(b)$ or the waiting time has been assumed.

Partition possible crossing times into $A_s(r, n) < D_r/(3\delta)$ and, for $j \geq 1$,

$$2^{j-1}D_r/(3\delta) \leq A_s(r, n) < 2^j D_r/(3\delta).$$

Apply (16) with $(b, z) = (D_r/(3\delta), D_r)$ for the first range and $(b, z) = (2^j D_r/(3\delta), 2^{j-1} D_r)$ for the others. Because $D_r \geq \alpha_r$, the respective bounds are at most

$$4V(1 + (3\delta)^{-1})\frac{\alpha_r}{D_r}, \quad 16V(1 + (3\delta)^{-1})\frac{\alpha_r}{D_r}2^{-j}.$$

Summing gives the first term in (13).

After reaching δ . Let $\rho = \inf\{n \geq r : D_n \geq \delta\}$, using $\rho = r$ if $D_r \geq \delta$. On $\{\rho < \infty, \rho \leq \tau\}$, restart at ρ and stop at $\eta = \tau \wedge \inf\{n \geq \rho : D_n \leq 0\}$:

$$\widetilde{M}_n = \sum_{k=\rho}^{n-1} \mathbf{1}_{\{k < \eta\}} Z_{k+1}.$$

For $D_k \geq \delta$, (12) and $0 \leq \alpha_k \sigma_k(s) \leq 1$ imply $D_k + m_k \geq \delta$. If a first subsequent crossing is at n , let $\ell < n$ be the last index with $D_\ell \geq \delta$. All subsequent gaps before n are in $(0, \delta)$ and have nonnegative mean increments, so

$$D_n \geq \delta + \widetilde{M}_n - \widetilde{M}_\ell.$$

A crossing forces $\sup_{n \geq \rho} |\widetilde{M}_n| \geq \delta/2$. Doob's inequality at ρ bounds this probability by $16V\delta^{-2} \sum_{k \geq \rho} \alpha_k^2$. Multiply by the $\mathcal{F}_\rho$ -measurable event $\{\rho < \infty, \rho \leq \tau\}$, average back to $\mathcal{F}_r$, and use $\rho \geq r$. This proves the second term. The last-visit index ℓ is used only in a pathwise inequality; we never condition at it. Both parts retain the increment $k = \tau - 1$ and therefore protect the gap through the endpoint itself. $\blacksquare$

3.4 Reference gaps and separately completed seeds

For a reference policy π , define

$$\begin{aligned} d_{\pi, s, a} &= q_\pi(s, \pi(s)) - q_\pi(s, a), & \mathcal{B}_\pi &= \{(s, a) : d_{\pi, s, a} > 0\}, \\ D_k^{\pi, s, a} &= q_k(s, \pi(s)) - q_k(s, a). \end{aligned} \tag{17}$$

An action in $\mathcal{B}_\pi$ would worsen the reference action according to q_π . If these estimated gaps are all positive, no greedy policy can select a bad action, so policy improvement applies. If every $\mathcal{B}_\pi$ is empty, improvement applies between every pair of policies in both directions. Their state values are equal; choosing a policy taking any prescribed action shows that every action attains that same value. All policies are optimal with the same target, and Lemma 7 completes the theorem. Otherwise set

$$d_* = \min_{\pi, (s, a) \in \mathcal{B}_\pi} d_{\pi, s, a} > 0, \quad \delta = d_*/4, \quad h = g = \delta/2. \tag{18}$$

Fix deterministic $K > B$ and restart r , and let $\zeta_r = \inf\{k \geq r : \|q_k\|_\infty > K\}$. For a stopping time $t \geq r$, work on $\{t < \infty, t < \zeta_r\}$, fix $\pi = \pi_t$, and define

$$\tau = \inf\{k > t : q_{\pi_k} \neq q_\pi\}, \quad \chi = \tau \wedge \zeta_r. \tag{19}$$

These are stopping times. Return moments give finite second moments for every fixed-time table by induction in (5). For $t \leq k < \chi$, the action-value target is exactly q_π , even if the selected policy label is different. Equal initial action probabilities give

$$\mathbb{E}[D_{k+1}^{\pi,s,a} - D_k^{\pi,s,a} \mid \mathcal{F}_k] = \alpha_k \sigma_k(s)(d_{\pi,s,a} - D_k^{\pi,s,a}), \quad (20)$$

$$\text{Var}(D_{k+1}^{\pi,s,a} - D_k^{\pi,s,a} \mid \mathcal{F}_k) \leq V \alpha_k^2 \sigma_k(s), \quad V = 2\{c_v + (B + K)^2\}. \quad (21)$$

Only the two coordinates forming the gap can change, their selection events are disjoint, and each contributes at most $\alpha_k^2 \sigma_k(s)\{c_v + (B + K)^2\}$ to the raw second moment. Since $d_{\pi,s,a} \geq 4\delta$, the state-clock barrier applies up to and including χ . All such moment estimates are used before exit, not conditioned on future containment in the ball.

A favorable local update. Suppose the reference action is selected at s and a current bad gap is below δ . Request one of the following action/return events:

1. If $q_\pi(s, \pi(s)) - q_k(s, \pi(s)) \geq d_*/2$, select $\pi(s)$ and require $G_{k+1} \geq q_\pi(s, \pi(s)) - h$. The reference coordinate increases by at least $3\delta\alpha_k/2$.
2. Otherwise select a and require $G_{k+1} \leq q_\pi(s, a) + h$. The identity

$$q_k(s, a) - q_\pi(s, a) = q_k(s, \pi(s)) - q_\pi(s, \pi(s)) + d_{\pi,s,a} - D_k^{\pi,s,a} > \delta$$

implies that the bad coordinate decreases by more than $g\alpha_k = \delta\alpha_k/2$.

In either case no other gap against the reference action decreases, and the reference action remains greedy. For centered X with variance at most c_v , the one-sided Chebyshev inequality gives

$$\mathbb{P}(X \geq -h) \geq \frac{h^2}{c_v + h^2}, \quad \mathbb{P}(X \leq h) \geq \frac{h^2}{c_v + h^2}.$$

For example, apply Markov's inequality to $(X + c_v/h)^2$ to bound $\mathbb{P}(X \geq h)$ and repeat for $-X$; when $c_v = 0$, $X = 0$ almost surely. Conditional on selecting s , its initial action is uniform, so every requested local update has conditional probability at least

$$p_0 = \frac{h^2}{\max_s |\mathcal{A}(s)| (c_v + h^2)} > 0. \quad (22)$$

No probability of selecting the state enters this bound.

Lemma 9 (Separately completed statewise seeds) *For fixed-priority or full-inertias and any positive integer N , there is an adapted construction asking for at most JN favorable local updates, starting at $t \geq k_m$. Along successful requests, every unfinished state retains its reference action, including at an endpoint reached by a favorable update. If state s completes at time $T_s < \infty$, then*

$$D_{T_s}^{\pi,s,a} \geq \min\{\delta, gN\alpha_{T_s}\} \quad ((s, a) \in \mathcal{B}_\pi). \quad (23)$$

Each request, conditional on selecting its unfinished state, has probability at least p_0 .

Proof. Order the bad actions at each state. Skip an action immediately if its gap is already at least δ ; otherwise process it with at most N favorable updates at that state, stopping earlier if it reaches δ . Process skips at t and after every favorable update, before the next sample. A state completes when its list is exhausted. Write T_s for this time, using $T_s = t$ for a state completed initially and $T_s = \infty$ if the construction stops before completion. These are stopping times. Every selection of an unfinished state requests a local update; completed states receive ordinary updates. Stop the construction at the first failed requirement or at χ .

Under full inertia, raising the reference estimate or lowering a competitor retains the reference action, since it remains greedy. At an unupdated state the table is unchanged, so the action is also retained. Induction proves the assertion through every unfinished state's wait.

For fixed priorities, at time t the reference action has highest priority among that state's maximizers. Leaving the table unchanged, raising its estimate, or lowering a competitor cannot create a new maximizer ahead of it in the priority order. Therefore that same action continues to be selected at every unfinished state. Induct over all iterations, including those updating other states and the final favorable update at completion. This proves the required retention property directly; a fixed-priority rule is not being identified with full inertia on arbitrary updates.

Every processed gap either reaches δ or receives N increases of at least $g\alpha_k$. Later seed updates at its state cannot lower it. Eventual nonincrease, already in force, gives $\alpha_k \geq \alpha_{T_s}$ for every such update and proves (23). At most J bad actions are processed, so there are at most JN requests, irrespective of waiting times. Their conditional probability bound is (22). $\blacksquare$

Completed states can be updated while other states wait. We therefore cannot condition indiscriminately on all future successful behavior: the completed-state return laws must stay unchanged. The following bounded change of measure conditions only the seed updates.

Lemma 10 (Conditioning only the seed updates) *There is an auxiliary law $\widehat{\mathbb{P}} \ll \mathbb{P}$ under which all active requirements of Lemma 9 succeed. Conditional state selection is unchanged. At completed states, and after the construction stops, the conditional action and return law given the current history and selected state is unchanged. For every future event E , on $\{t < \infty, t < \zeta_r\}$,*

$$\mathbb{P}(E \mid \mathcal{F}_t) \geq p_0^{JN} \widehat{\mathbb{P}}(E \mid \mathcal{F}_t). \quad (24)$$

Proof. Let S_{k+1} be the state component of I_{k+1} . At an active unfinished state s , let $p_k(s)$ be the conditional probability of its requirement given $\mathcal{F}_k, S_{k+1} = s$. Then $p_k(s) \geq p_0$. If the state has zero sampling probability, set $p_k(s) = p_0$; this value is never used on a positive-probability draw.

Start $L_t = 1$. At a requested update of s , multiply L by the indicator of success divided by $p_k(s)$; at other iterations multiply it by one. Every factor has conditional mean one given $\mathcal{F}_k, S_{k+1}$. Hence (L_k) is a martingale. At most JN factors differ from one, even though their times need not be bounded, so $0 \leq L_k \leq p_0^{-JN}$. Bounded convergence gives $\mathbb{E}[L_\infty \mid \mathcal{F}_t] = 1$. Define $d\widehat{\mathbb{P}} = L_\infty d\mathbb{P}$. For random t , use likelihood one before t , on $\{t = \infty\}$, and off $\{t < \zeta_r\}$, and combine the constructions on $\{t = j\}$ over integers j . The law agrees with $\mathbb{P}$ on $\mathcal{F}_t$ where the construction is active.

A failure makes $L_\infty = 0$, so active requirements succeed almost surely under the auxiliary law. To check conditional laws, use $\mathbb{E}[L_\infty \mid \mathcal{F}_{k+1}] = L_{k+1}$; future factors do not alter the one-step likelihood. Its current factor has conditional mean one given the selected state, preserving state selection. At a completed state the current factor is identically one, so its conditional action and return law is unchanged as well. The statement holds on auxiliary-almost-every history, where $L_k > 0$. Absolute continuity preserves the almost-sure algorithmic identities, including the chosen fixed-priority or full-inertia tie rule. Finally,

$$\widehat{\mathbb{P}}(E \mid \mathcal{F}_t) = \mathbb{E}[L_\infty \mathbf{1}_E \mid \mathcal{F}_t] \leq p_0^{-JN} \mathbb{P}(E \mid \mathcal{F}_t),$$

which proves the bound. This auxiliary law is used only in the proof; the algorithm still samples its original returns. $\blacksquare$

3.5 A stopped positive probability of progress

Let $W(\pi) = \sum_{i \in \mathcal{I}} q_\pi(i)$. Its possible values form a finite ordered set. A distinct componentwise larger target has strictly larger W , although different targets may have equal W .

Lemma 11 (Stopped improvement bound) *Under the assumptions of Theorem 4, for every fixed $K > B$ there are deterministic k_0 and $p > 0$ such that for every deterministic restart r and stopping time $t \geq \max\{r, k_0\}$, on $\{t < \infty, t < \zeta_r\}$,*

$$\mathbb{P}\left(\left\{\zeta_r \leq \tau, \zeta_r < \infty\right\} \text{ or } \left\{\tau = \zeta_r = \infty\right\} \text{ or } \left\{\tau < \zeta_r, W(\pi_\tau) > W(\pi_t)\right\} \mid \mathcal{F}_t\right) \geq p. \quad (25)$$

Here τ is the next target change in (19).

Proof. Fix V from (21) and C_0, C_1 from the state-clock barrier. Choose a positive integer N with $2JC_0/(\delta N) \leq 1/4$. Then choose deterministic $k_0 \geq k_m$ so that, for every $t \geq k_0$,

$$N\alpha_t \leq 2, \quad \frac{JC_1}{\delta^2} \sum_{k=t}^{\infty} \alpha_k^2 \leq \frac{1}{4}. \quad (26)$$

The radius and variance constants are fixed before choosing N .

Use the auxiliary law of Lemma 10. For each bad action whose state completes at $T_s < \chi$, apply the barrier at T_s , stopped at χ . At that completed state the original sampling and return laws are preserved, so (20)–(21) hold under $\widehat{\mathbb{P}}$. No such assertion is made for a conditioned return at an unfinished state. Nonincrease and (23)–(26) imply that its conditional crossing probability is at most

$$\frac{2C_0}{\delta N} + \frac{C_1}{\delta^2} \sum_{k=t}^{\infty} \alpha_k^2. \quad (27)$$

Indeed $gN\alpha_{T_s} \leq \delta$, so $\alpha_{T_s} / \min\{D_{T_s}^{\pi, s, a}, \delta\} \leq 1/(gN)$. Multiply the conditional estimate by the $\mathcal{F}_{T_s}$ -measurable event $\{T_s < \chi\}$ and average back to $\mathcal{F}_t$. A union bound over at most

J pairs gives probability at most $1/2$ of any completed gap failing through χ . This needs neither independence nor simultaneous completion. If $T_s = \chi < \infty$, its final favorable update already retains the reference action, so no subsequent barrier is needed.

On the complementary event, either exit occurs first, the target persists without exit, or $\tau < \zeta_r$ is finite. In the third case, states unfinished just before the endpoint retain the reference action, including those completed at the endpoint. Previously completed states cannot choose a bad action. Thus

$$q_\pi(s, \pi_\tau(s)) \geq q_\pi(s, \pi(s)) \quad \text{for every } s.$$

Policy improvement gives $q_{\pi_\tau} \geq q_\pi$ with distinct targets, and hence a strict increase of W . There is no need to prove that every seed completes: at a finite endpoint unfinished states are safe, and an infinite endpoint without exit is already a successful outcome. Transfer back by (24), obtaining (25) with $p = \frac{1}{2}p_0^{JN}$.

If $\mathcal{B}_\pi$ is empty, the gap conditions and barrier are vacuous and policy improvement holds for every next selected policy. The same lower bound remains valid. An exit *after* an unsuccessful finite target change is not counted as success for that earlier interval. ■

3.6 Finitely many target changes and convergence

Proof of Theorem 4. Fix $K > B$ and constants k_0, p from Lemma 11. For a deterministic $r \geq k_0$, start at $t_0 = r$ and absorb immediately if $\|q_r\|_\infty > K$. Enumerate successive target changes

$$t_{n+1} = \inf\{k > t_n : q_{\pi_k} \neq q_{\pi_{t_n}}\}$$

only while $t_n < \zeta_r$. All finite continuing times are stopping times. Stop counting if exit occurs before or at the next target change, or if the target persists forever without exit.

At every alive interval, absorption or a strict increase in W has conditional probability at least p . By induction and the tower property, absorption within the next m intervals or m consecutive strict increases has conditional probability at least p^m . On an absorbed branch the continuation probability is one; on a finite successful-transition branch condition at its stopping time and apply the remaining bound. This uses no independence and never conditions at an infinite time on a continuing branch.

Let L be the number of distinct values of W . There cannot be L consecutive strict increases. Absorption within the next L intervals therefore has conditional probability at least p^L . Applying the bound at successive surviving blocks gives probability at most $(1 - p^L)^j$ of surviving j such blocks. In particular the probability of infinitely many target changes before exit is zero.

Intersect these probability-one conclusions over all deterministic integers $r \geq k_0$. Lemma 7 says that almost every path remains inside the ball after some such integer. On that path, choose r with $\zeta_r = \infty$; the original sequence consequently has only finitely many target changes. The integer is selected pathwise only after the countable intersection. We do not condition on an eventual containment time or claim that it is a stopping time. Nor is an unconditional geometric tail for the original process's total number of changes asserted.

Lemma 7 now gives $q_k \rightarrow q^*$. Every nonoptimal action has a strictly positive optimality gap. Finiteness of the action sets and convergence exclude all such actions from sufficiently

late greedy policies. Those policies are greedy for q^* and hence optimal by Bellman verification. If every state has a unique optimal action, their labels are eventually constant. ■